

Module: tf.dtypes

Public API for tf._api.v2.dtypes namespace

Modules

experimental (https://www.tensorflow.org/api_docs/python/tf/dtypes/experimental) module: Public API for tf._api.v2.dtypes.experimental namespace

Classes

class DType (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType): Represents the type of the elements in a **Tensor**.

Functions

as_dtype(...) (https://www.tensorflow.org/api_docs/python/tf/dtypes/as_dtype): Converts the given **type_value** to a **tf.DType** (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType).

cast(...) (https://www.tensorflow.org/api_docs/python/tf/cast): Casts a tensor to a new type.

complex(...) (https://www.tensorflow.org/api_docs/python/tf/dtypes/complex): Converts two real numbers to a complex number.

saturate_cast(...) (https://www.tensorflow.org/api_docs/python/tf/dtypes/saturate_cast): Performs a safe saturating cast of **value** to **dtype**.

Other Members	
QUANTIZED_DTYPES	<pre>{ tf.qint16, tf.qint16_ref, tf.qint32, tf.qint32_ref, tf.qint8, tf.qint8_ref, tf.quint16, tf.quint16_ref, tf.quint8, tf.quint8_ref }</pre>
bfloat16	Instance of tf.dtypes.DType (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 16-bit bfloat (brain floating point).
bool	Instance of tf.dtypes.DType (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Boolean.
complex128	Instance of tf.dtypes.DType (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 128-bit complex.
complex64	Instance of tf.dtypes.DType (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 64-bit complex.
double	Instance of tf.dtypes.DType (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType)

	64-bit (double precision) floating-point.
float16	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 16-bit (half precision) floating-point.
float32	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 32-bit (single precision) floating-point.
float64	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 64-bit (double precision) floating-point.
half	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) 16-bit (half precision) floating-point.
int16	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Signed 16-bit integer.
int32	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Signed 32-bit integer.
int64	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Signed 64-bit integer.
int8	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Signed 8-bit integer.
qint16	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Signed quantized 16-bit integer.
qint32	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) signed quantized 32-bit integer.
qint8	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Signed quantized 8-bit integer.
quint16	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Unsigned quantized 16-bit integer.
quint8	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Unsigned quantized 8-bit integer.
resource	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Handle to a mutable, dynamically allocated resource.
string	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Variable-length string, represented as byte array.
uint16	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Unsigned 16-bit (word) integer.
uint32	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Unsigned 32-bit (dword) integer.
uint64	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Unsigned 64-bit (qword) integer.
uint8	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType) Unsigned 8-bit (byte) integer.
variant	Instance of <code>tf.dtypes.DType</code> (https://www.tensorflow.org/api_docs/python/tf/dtypes/DType)

Data of arbitrary type (known at runtime).

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